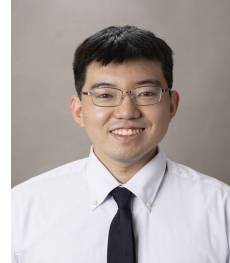


Restaurant AI Checkpoint 3

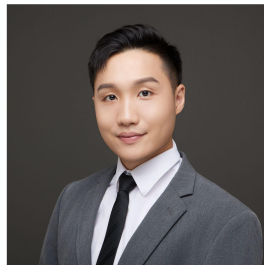
Growing interest in local entrepreneurship—especially in high-risk sectors like restaurants—combined with limited guidance leads to costly mistakes, but AI can integrate market analysis, financial modeling, and compliance checks to reduce risk and lower barriers to entry.



Mubtasim Raad



Takumi Nishi



Alex Chen



Laura Shih



<https://github.com/IS492-SP26/Restaurants-AI/tree/main>

Final Problem Statment



Problem statements

- Starting a restaurant or local business is overwhelming for first-time entrepreneurs.
- It's challenging to navigate market validation, financial planning, compliance, and operational risks.
- The information current AI tools generate are often without structured guidance and verification.

How restaurant AI address them

- Users are able to personalize and customize the entire business planning process from minimal inputs.
- Users are able to obtain a comprehensive business plan across marketing, financial, regulatory, and operational analysis with multi-agentic workflow.
- The restaurant AI tools provide users feasible business plans with regulatory and compliance verification.
- Multi-agentic workflow with an “All in One” interface.



Final Problem Statement



Problem Statements

- Starting a restaurant or local business is overwhelming for first-time entrepreneurs.
- It's challenging to navigate market validation, financial planning, compliance, and operational risks.
- The information current AI tools generate are often without structured guidance and verification.

Our AI Solution

- Personalized business planning from minimal inputs.
- Comprehensive multi-agentic workflow across marketing, finance, regulation and operations.
- Feasible business plans with built-in regulatory and compliance verification.
- "All-in-One" interface for seamless navigation.



Target User

- ✓ First-time entrepreneurs
- ✓ Students, and career switchers starting small businesses.
- ✓ Passion-driven founders (chefs, home cooks) without formal business training.
- ✓ Passionate Individuals without access to consultants, lawyers, or startup capital



Live Demo



Design Decisions & Challenges



Agents	AI Models Used & Why	Data Collected	Challenges
Regulatory (Raad)	Claude, GPT-4o-mini (via OpenAI API). Cost effective at roughly \$0.001 per question, Provides better performance.	All public data including: Champaign, IL code of ordinances, food ordinances, zoning ordinance, Illinois food service Sanitation Code, Illinois Food Complinance and Enforcement.	Initially used Groq free tier has rate limits causing errors during heavy testing.
Operations (Alex)	Claude, GPT-4o-mini (via OpenAI API)	Operational data including labor and workflow related data, operational know-how and quality management guidance	Ensuring the model outputs are structured and feasible guidance
Financial (Laura)	Cursor, GPT-4o -mini via OpenAI API, then fine-tuned for financial planning. Chosen because it is fast, lower cost, supports structured JSON, and gives more consistent budget and cash-flow outputs.	Public financial data including CPI-U inflation, Consumer Expenditure Survey income data, FRB H.15 interest rates, Illinois county wage data, BLS weekly earnings, and startup cost worksheet categories.	A key challenge was deciding which financial data is most important to feed into the agent for training, such as inflation, wages, interest rates, consumer spending, and startup cost categories.
Marketing (Takumi)	Claude, GPT-4o-mini (via OpenAI API)	Data relating to marketing strategies (research and branding), Restaurant information (address, type) in Champaign	Structure of output for marketing information



Safety, Privacy & Reliability



Data Privacy (No PII)

The application protects privacy by collecting only business-related info. No personal identifying information (PII) or sensitive data is stored.

Regulatory Compliance

Outputs are verifiable, citing real-world sources like CUPHD and FDA Food Codes, allowing users to cross-check guidance.

User Disclaimer

Defined as an educational planning assistant. Users are directed to consult professionals for legal or financial advice.

API Key Security

API keys are excluded from the public Github repo. Credentials are never exposed.

Educational planning assistant. Not legal, tax, or financial advice. Local rules vary—verify permits, leases, and licenses with qualified professionals.

MVP uses deterministic mock agents; outputs are illustrative.



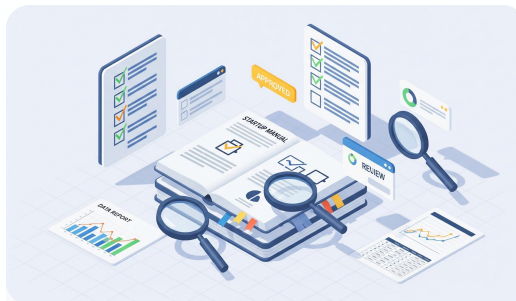
Risks, Safeguards and Next step (CP3)

Next Milestones for CP4



User testing

Recruit entrepreneurs and students to test the 3-step flow (Basics → Ideas → Manual) and measure completion time, usefulness, and trust.



Evaluate output quality

Assess if generated manuals offer actionable, accurate guidance compared to general-purpose AI tools across various business domains.



Address Hallucination risk

Tackle hallucination risk by implementing confidence scores and verification layers to ensure high reliability of generated content.